

Virtual Reality as Philosophy Machine: A Proof-of-Concept Study

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Abstract

Philosophical reflection involves deep, critical engagement with fundamental questions about reality and can support both individual self-awareness and broader critical thinking. Yet contemporary philosophical education faces growing challenges due to the fast pace of life and the prioritization of “economically useful” competencies. This study examines the educational potential of VR to foster philosophical reflection among students. We developed a VR application, [anonymized], and conducted a proof-of-concept qualitative study with 56 participants using individual questionnaires and semi-structured focus group interviews, followed by thematic analysis. Findings suggest that VR—through first-person perspective, multisensory experience, and heightened engagement—can effectively support philosophical self-reflexivity, while distractors such as technical issues, novelty effects, and time pressure may hinder reflectiveness. We propose design and research recommendations and argue that well-designed VR experiences may function as a “philosophy machine” for supporting philosophical reflection in digital society.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

virtual reality, VR, philosophy, education, critical thinking, proof-of-concept study.

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1 Introduction

Virtual reality (VR) has become the subject of growing debate about its capacity to reshape experience, social interaction, and perceptions of reality. Beyond entertainment, research increasingly highlights its cognitive and educational potential [25, 29]. Barbot and Kaufman [3] suggest that, when designed well, VR can function as an “empathy machine.” Building on this idea, we ask whether VR could also operate as a “philosophy machine,” supporting philosophical reflection—deep engagement with questions of existence, values, reality, and identity.

Philosophical education has documented benefits, including improved creativity and language skills [19], strengthened critical thinking in children [38, 40], and greater resilience in adults through philosophical counseling [36]. Yet the cultivation of philosophical thinking is increasingly constrained by economically pragmatic educational agendas [5, 27], accelerated life rhythms that undermine deep focus [30], and declining engagement with complex texts [10, 37]. VR may respond to these pressures by increasing engagement, supporting concentration, and making abstract ideas more tangible through immersion.

VR can also be framed as a philosophical laboratory: Plato’s cave allegory (*Republic*) concerns the instability of perceived reality, a theme that resonates with VR as a medium that simulates worlds while inviting epistemic doubt and reflection. This proof-of-concept study [11] examined whether VR can foster philosophical reflection. Fifty-six participants experienced [anonymized], an interactive VR adaptation of Plato’s allegory developed by the project team. We analyzed the presence and character of reflections elicited by the experience (including personalization, internalization, and subjective value) and identified factors that supported or hindered reflectiveness, spanning design, user experience, and environmental context. The study addressed three questions: (1) to what extent VR can foster philosophical reflection, (2) how students assess this potential, and (3) which aspects of the experience create barriers or opportunities for reflectiveness.

2 Related Studies

VR in education has been widely studied, with systematic reviews indicating benefits for knowledge acquisition, skills training, and the cultivation of pro-social and pro-environmental attitudes [16, 17, 22, 25]. It is also increasingly used for history education and cultural heritage dissemination [28, 39]. By contrast, the use of VR specifically to foster philosophical reflection remains largely underexplored.

Recent evidence also suggests that VR can raise awareness of socio-cultural issues and support attitudinal change. Meta-analytic work indicates that higher levels of immersion tend to produce stronger effects than print, 2D video, or oral narration [25]. Prior research further argues that users may internalize virtual experiences and treat them as meaningful, which can make otherwise abstract problems (e.g., homelessness, climate change) experientially salient [1, 21]. Perspective-taking in VR has been linked to reduced prejudice and greater willingness to act on behalf of others, while immersive exposure can increase interest, learning, and—compared with text-based messaging—support behavior change by narrowing the knowledge-to-action gap [1, 15, 21, 25, 35].

These effects are often interpreted through embodied cognition: learning is strengthened when it is situated, sensorimotor, and experienced from a first-person perspective [13, 14, 18, 21, 31]. In VR, multisensory engagement and bodily action can scaffold meaning-making and facilitate learning [18]. Building on this literature, we ask whether VR's embodied and immersive affordances can support the internalization of, and engagement with, philosophical reflection.

Existing research on VR and philosophical thinking has thus far focused primarily on simulating moral dilemmas (e.g., the trolley problem) [12, 24, 32]. These studies typically compare decision-making in VR with other experimental formats rather than aiming to cultivate philosophical reflection in educational settings. An exception is Collette's [8] pedagogical study, which describes the use of a commercially available VR application simulating the experience of walking on a plank suspended over an abyss, employed to enhance student engagement with existentialist philosophy. Lee & Lee [20] describe the design process of the Sunghaksipdo VR application intended to present conceptual heritage connected to Neo-Confucian philosophy. Although the authors share similar assumptions to ours in the design process, attempting to utilize embodied interactions as a means of constructing meaning, they do not provide a methodological evaluation regarding the app's effectiveness in fostering philosophical reflection. There is a notable lack of research addressing the potential of VR as a tool for exploring complex, abstract philosophical concepts – such as Plato's cave metaphor – making this field highly underexplored.

3 [anonymized] – Concept and Implementation

The application was developed through an interdisciplinary collaboration within an experimental VR initiative that adapts classical philosophical ideas. It is intended to complement text-based philosophy education and engage audiences beyond academia. Rather than illustrating Plato's text literally, the experience translates its core claims into a multisensory VR environment, using embodied interactions (e.g., grasping, propulsion, and navigation) to intertwine

physical effort with reflective work. In doing so, it foregrounds the relationship between thinking and bodily activity and problematizes a strict mind–body dualism discussed in embodied cognition scholarship.

The experience begins in a submerged cave where the user is restrained and must free themselves with the help of nonhuman guides. They move through corridors while hearing fragments of philosophical speech and film dialogue that allude to the cave metaphor, encountering objects that reveal themselves as unstable or illusory. The aesthetic is deliberately surreal. After surfacing, the user is lifted toward a shaft of light—only for the scene to destabilize: spatial orientation reverses and familiar distinctions (up/down, end/beginning) collapse. Seeking an exit, the user returns to the cave they left, but it has transformed. At its far end, they find a virtual headset; putting it on transports them into an abstract, geometric space. A final prompt suggests removing first the virtual headset and then the physical one, leaving the user with the unresolved question of whether any “ultimate” exit is possible.

The project has been presented publicly in international and local contexts, including curated showcases focused on immersive art and events at the intersection of digital culture and academic exchange. The application was built with a widely used real-time 3D engine and a standard VR runtime. The study reported in this article relied on the version compatible with broadly available consumer hardware.

Development involved substantial conceptual and technical challenges. Conceptually, the adaptation was guided by the premise that the intensity of the original philosophical drama can be conveyed by translating intellectual struggle into bodily strain. As a result, a measured degree of discomfort was treated as a design feature rather than a defect. However, calibrating the balance between effort and satisfaction proved difficult, and late-stage revisions required major restructuring of the scenario, including removing a significant portion of previously completed content.

Another challenge was keeping users' attention on the metaphor's philosophical stakes rather than on dexterity-oriented tasks. Conventions associated with exploratory VR experiences—complex locomotion and “finding the way out”—risked shifting the experience toward puzzle-solving. In response, the scenario incorporated scripted moments (via audiovisual and narrative cues such as surprise and disorientation) designed to prompt pauses for reflection, alongside a light-touch guiding voice in the early segment. Locomotion tuning posed persistent issues: more realistic mappings increased immersion but slowed progress and discouraged exploration, while more dynamic mappings improved control at the cost of higher collision risk. Additional traversal mechanics were ultimately simplified after repeated failures and edge cases (e.g., mode-switch interruptions and immobilization), and extensive testing was required to stabilize navigation and environmental collision behavior.

4 Methods

The study explored whether VR can foster philosophical reflection through a proof-of-concept design [11]. Fifty-six participants used a bespoke application ([anonymized]). To strengthen credibility, we applied triangulation across methods (open-ended questionnaires

and focus groups), investigators (three researchers independently moderated and analyzed data), and environments (two physical settings) [2, 4].

Participants experienced the application in either a neutral computer-room setting (n=37) or a curated installation (n=19) designed to evoke the cave metaphor. Each participant completed a 30-minute session including briefing, self-guided exploration, and an open-ended questionnaire; researchers intervened only when navigation support was needed. Sessions were conducted with an HTC VIVE Pro headset (2880×1600, 110° FOV) and Vive controllers, supported by a high-performance PC (Intel Core i9, 64 GB RAM, NVIDIA GeForce RTX 4080). One to two weeks later, four focus interviews (about 14 participants each) were conducted following established procedures [9], with one moderator per group (three facilitators in total). Participants were students from a single Polish university (Cultural Studies n=37; Social Informatics n=15; other programs n=4), mostly aged 20–23; approximately 30% identified as male and 70% as female (precise data were lost). Participation was voluntary and unpaid; ethics approval was obtained. Questionnaires included 11 open-ended items covering (1) user experience, (2) philosophical interpretation, and (3) perceived value. Focus groups addressed shared interpretation, facilitators and barriers to reflectiveness (e.g., environment, social context, technical issues), and perceived educational potential of VR; discussions were recorded, transcribed, anonymized, and translated.

Data were analyzed via inductive thematic analysis informed by grounded theory [7, 34]. Three researchers coded independently, compared themes mid-way and at completion to support reflexivity and trustworthiness [26], and contrasted code patterns between the two environment cohorts. Analysis was conducted in MAXQDA 24.

5 Results

RQ1 To what extent can VR be a tool for fostering philosophical reflection? Participants' reflections were often complex, emotionally charged, and positioned along a spectrum from detached commentary to highly personal engagement. Many moved beyond the application's literal content or Plato's text, linking the experience to their own lives and articulating individualized interpretations of universal themes: *The cave speaks to the limitations of our perception and how our way of seeing the world is shaped by our individual beliefs, experiences, and social contexts. It makes us realize that what we perceive as truth may only be our version of it.* Direct references to Plato were present but not dominant; some participants framed the experience metaphorically, e.g., as dreamlike or introspective: *Like Alice, by putting on the headset we jump into the rabbit hole. We move through an underwater Wonderland, encountering different situations, objects, and quotes. Then, we take the headset off and return to reality. Our adventures didn't happen in the real world, but we remember them like a dream.* Others described more personal, lasting engagement: *The moment when I was discovering new symbols and 'levels' of the cave was particularly meaningful to me. It felt like uncovering new layers of perception, which made me reflect on how limited our understanding of the world around us might be.* Reflections frequently continued after the session through extended discussion: *It's not just something printed on a*

page or in a book – you're in it. You can feel it in some way and then reflect on it even more. [...] I was with another person, and we talked about it [after the experience] for another hour, practically nonstop. We had different interpretations, different feelings, and we were just comparing our experiences. I doubt we would've had such a conversation if we had only read the text. Emotional responses were intense and layered—typically fear, fascination, and curiosity: *"It was an intriguing journey into my consciousness. The moment that stood out most was encountering the eye that followed my movements: it was both terrifying and exciting."* Environment mattered: reflections in the curated condition tended to be longer, more complex, and more personally involving, with stronger and more nuanced affect; the neutral-setting group more often reported detached reactions and simpler "curiosity." Nonetheless, complex reflections appeared for most participants.

RQ2 How do students assess the potential of VR technology to foster philosophical reflection? Most participants viewed VR as conducive to philosophical reflection because interactivity and embodiment make the experience personally involving. Unlike passive reception of texts or films, VR positions the user as protagonist and can intensify self-reflection: *"Because it is more personal – more of an actual experience – it's not merely something presented on a page or in a book; rather, you are in it,"* as one participant remarked. Participants highlighted three features that most supported engagement: (1) first-person perspective (*"Here, we are more ourselves – we do not think about what the character is doing, but rather about what we are doing"*); (2) immersion and perceived realism; and (3) multimodality, which deepens experiential impact: *"It affects many senses, affects us. We can truly get to know more, get to know ourselves, and reflect more deeply on these concepts."* VR was also described as helpful for making abstract problems more graspable through interactive experience (*"It is a tool that can really help, for example, with the problem of abstract thinking"*). At the same time, participants noted the difficulty of communicating intended meaning to broad audiences; notably, interpretive openness was sometimes seen as a strength that can itself provoke reflection: *"I think that, depending on the person, everyone can read it a bit differently [...]. After all, philosophy itself often allows us to think about these different ideas in a fairly open way, so that everyone can interpret it in their own way."* However, reflective engagement depends on pacing and user familiarity. Several participants argued that time pressure disrupts understanding: *"I felt pressured to keep moving forward; I didn't have time to focus on understanding. So here, I think it's important to consider how much we want to engage the user – to make them do something, but also to ensure there is time to understand the story at all."* They also emphasized that novelty can overshadow content, especially for first-time users: *"It depends on the user's prior experiences. For instance, if someone has already used VR before, they might simply be interested in what is happening around them and, in that context, want to understand the story. But if it is a user who is experiencing VR for the first time, they might be more fascinated by the fact that – wow – it's VR."* Focus groups further suggested that learning controls can consume attention and that gaming associations can encourage "completion" rather than contemplation: *"Someone might come – just as you said – just to try out VR, to go through the application, [...] just to see what's there, and that's all. But someone else might come to actually explore the*

content of the application [...]. Because there's a difference between simply going through the app [...] and that phase of philosophical reflection – those insights we arrive at on our own.”

RQ3 What factors of the VR experience in [anonymized] constitute challenges or potential opportunities for fostering philosophical reflectiveness among students? Although [anonymized] was engaging, participants identified constraints that can undermine contemplation: non-intuitive controls, problematic affordances, and texture errors produced frustration, disorientation, and occasionally discomfort, shifting attention away from meaning-making. These disruptions sometimes prompted a more passive stance, with the experience treated as a game rather than a contemplative encounter; the presence of others and cable management were additional distractors. Conversely, narrative treatments that interrupted exploration and redirected attention toward interpretation were well received, especially surreal elements (e.g., watching eyes, chains, flying fish): *I was surprised by the changes in the mechanics of the application, which appeared at unexpected moments (such as floating on a fish) and absurd insertions, such as a toilet appearing with an ironic comment that it's not a toilet. These moments added distance and surprising levity to the philosophical nature of the exhibition, creating an interesting mix of reflection and humour.* Participants also reported that surrealism enabled polysemic, personally grounded interpretation, reinforcing immersion while prompting introspection on perception, technology, and symbolic meaning: *A particular moment, I think important plot-wise in the whole application, was the moment when the whole location was turned upside down. It was important to me because of the symbolism of the prison. I also paid attention to the reflected shadow of the fish, which was meant to allude to all the philosophy presented in Plato's Cave. An additional element that interested me was the large eye among the chains. It monitored our movements and watched us, which also has its meaning in all the symbolism.* Intertextuality was likewise valued: even when references were not explicitly recognized, they encouraged deeper exploration of ideas, active meaning-seeking, and a more layered experience; one respondent described “easter egg” hunting as intellectually engaging: *“In addition, I found the symbolic elements, such as Greek letters, numbers and hands appearing, intriguing, and looking for ‘easter eggs’ was a form of intellectual entertainment for me.”*

6 Discussion

This proof-of-concept study suggests that VR can meaningfully support philosophical reflectiveness, extending prior work on VR in education [17]. While engaging with [anonymized], many participants articulated reflections on reality, perception, and truth, frequently relating these themes to their own lived experience. They also evaluated VR as a promising medium for philosophical inquiry: immersion, multimodality, and first-person perspective were described as producing a sense of involvement and protagonism that may stimulate reflection more than text-based engagement.

At the same time, reported barriers largely align with recurrent challenges in VR design [6]. Non-intuitive affordances, environmental imperfections, and related disorientation or frustration—sometimes accompanied by physical discomfort—can increase cognitive load and divert attention from meaning-making. In our

context, additional distractors such as time pressure, physical constraints of the setup (e.g., cables, instructor presence), technical glitches, and novelty effects commonly observed in educational VR [23] appear particularly detrimental to sustained philosophical reflection.

Philosophical VR also poses design-specific tensions. Because users vary in their familiarity with and interest in philosophy, content must remain accessible and non-intrusive, while still offering depth. This requires calibrating cultural/philosophical references (including “easter eggs”) so the experience remains coherent even for users who do not recognize them, and managing narrative pacing to create space for reflection during and after use. Finally, designers must contend with the dominant framing of VR as entertainment; positioning an experience as contemplative (“philosophy machine”) can shape expectations and requires careful priming and communication of intent.

Based on these findings, we propose practical recommendations for VR-as-philosophy-machine design: (1) provide brief pre-use training to reduce novelty and navigation-related cognitive load; (2) prioritize technical robustness and comfort to minimize physical and attentional disruption; (3) regulate pacing and incorporate intentional pauses, alongside a post-experience setting that supports quiet contemplation; (4) embed prompts and facilitate follow-up discussion to consolidate reflective work; and (5) adapt philosophical texts through medium-specific artistic reinterpretation rather than literal translation, leveraging embodied interaction and experiential dramaturgy to provoke inquiry.

As an exploratory qualitative study [33], our evaluation is limited in replicability and susceptible to interpretive subjectivity, especially because some researchers were involved in the application's design. We mitigated this risk through multi-level triangulation, including investigator, methodological, and environmental triangulation [2]. Generalizability is also constrained by focusing on a single application and a culturally specific participant cohort; additional philosophical VR applications and more diverse samples are needed to test the robustness of our observations.

Future work should expand to other philosophical concepts to determine which topics and forms of reflection are particularly well-suited to VR. Comparative studies—combining qualitative analysis with standardized questionnaires—should evaluate VR experiences against text-based counterparts. Finally, longitudinal designs are needed to assess longer-term effects on philosophical thinking and to examine how prior VR familiarity moderates users' capacity for contemplative engagement in virtual environments.

7 Conclusions

This proof-of-concept study suggests that VR can function as a “philosophy machine,” supporting philosophical reflection and expanding possibilities for philosophy education in digital societies. Future work should validate these early findings through comparative and longitudinal designs, including quantitative measures. Advancing this agenda will require sustained collaboration across art, philosophy, psychology, and HCI to design VR experiences that reliably support critical, meaning-oriented reflection at scale.

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A VRsophy: The Cave - Photos

The appendix contains a set of representative screenshots from the VR experience to visually contextualize the environment, interactions, and design elements referenced in the paper.